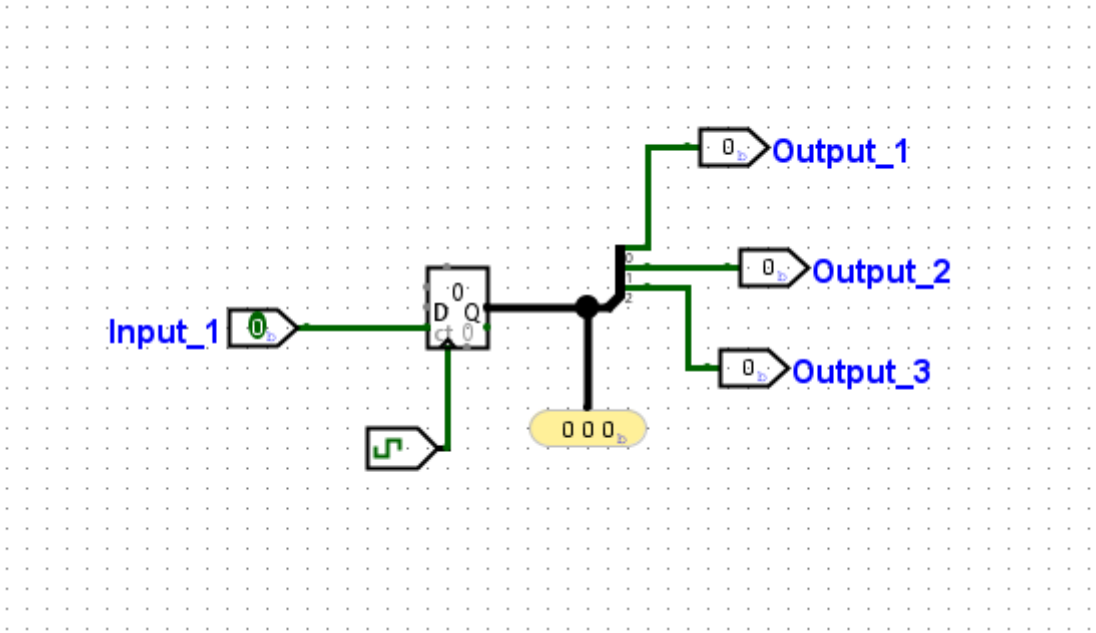
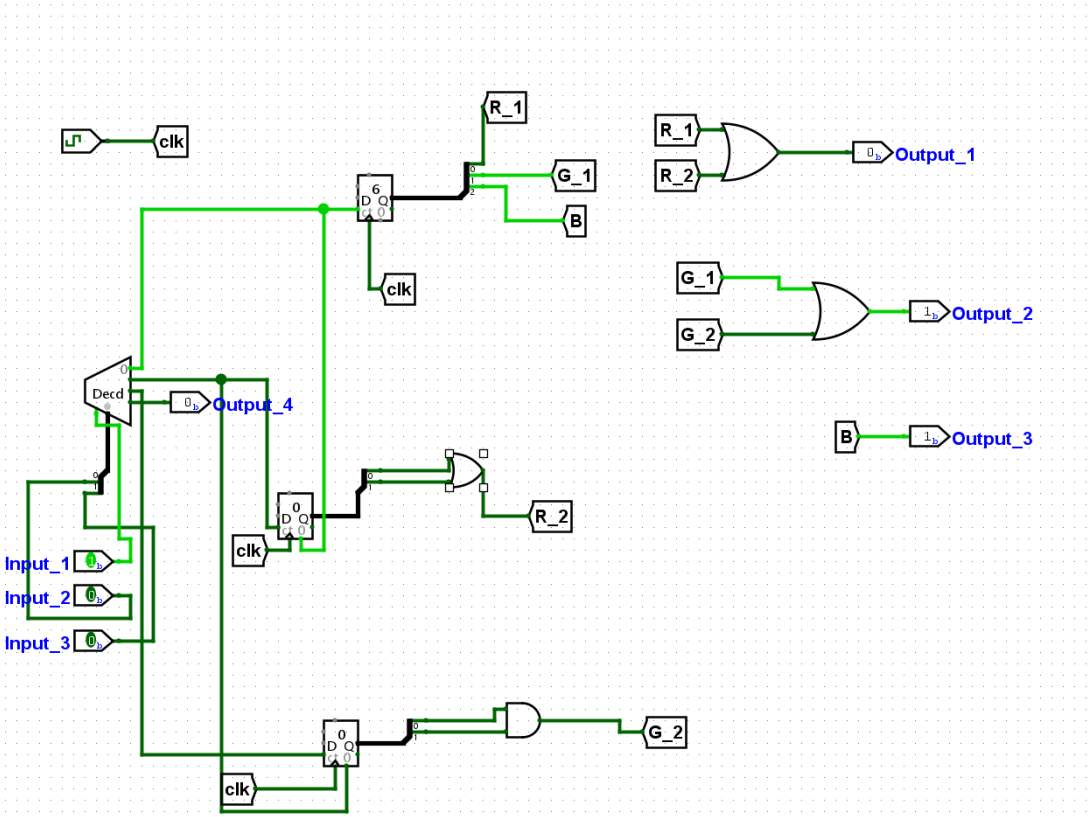
# Lab 8

# 3bit color cycler



# RGB LED Cycle switcher

- Uses decoder to switch pattern
- Uses counters to adjust duty cycle of clk for different rates
- Had Mind boggling error between Logisim and quartus, turns out it was the difference between a “ and a ‘



# Source of my suffering

```
MORY_5 : LogisimCounter
 GENERIC MAP ( invertClock => 0,
               maxVal      => "11",
               mode        => 0,
               width       => 2 )
 PORT MAP ( clear      => s_logisimNet3,
            clock      => logisimClockTree0(4),
            compareOut => OPEN,
            countValue => s_logisimBus6(1 DOWNTO
            enable     => s_logisimNet4,
            load       => '0',
            loadData   => "00",
            tick       => logisimClockTree0(2),
            upNotDown  => '0' );
```

- The max value was set to ‘1’ previously, when quartus expected a “1”
- Apostrophes reference a single bit, quotes reference a bus
- Since the counter was a single bit, Logisim set it as an apostrophe. This was solved by making the counter 2 bits